Augmented: the clean table

One row per element, 20 plain columns in standard units, built once when a model is indexed (augmented/clean.js) · shown here: rstbasicsampleproject.rvt, 845 elements

Before → after: one steel column

What Autodesk sends (text pairs)
`Constraints.Base Level` = "Level 2"
`Dimensions.Volume` = "0.010 m^3"
`Element.Type` = "200UB25.4"
`Materials and Finishes.Structural Material` = "Metal - Steel - 345 MPa"
`Structural Analysis.Nominal Weight` = "25.400 kgf/m"
`Structural Section Geometry.Height` = "20.300 cm"
`Structural Section Geometry.Width` = "13.300 cm"
...plus ~90 other properties

→

elements_clean
type **200UB25.4** · level **Level 2**
material_class **steel** · is_structural **1**
width_mm **133** · depth_mm **203** (from cm)
section **133 x 203 mm**
weight_kg_per_m **25.4** (from kgf/m)
height_m **3.1**⚙ (from the 3D view)
volume_m3 **0.01**

```
SELECT * FROM elements_clean WHERE category = 'Structural Columns' AND width_mm > 300 → 1 (the 450 mm concrete column)
```

What the cleaning does

- **units per value** cm, mm, ft-in, kgf/m, cm³ → mm / m / m² / m³ / kg / kg/m
- **text stays text** "200UB25.4", "88.9x5.0CHS", OmniClass codes are not numbers
- **placeholders** "<By Category>", "None", "-" → empty (= not set)
- **Revit rulebook** wall Width = thickness · square column b = width and depth · rebar "B" is not a section · material in two places
- **computed ⚙** weight = kg/m × length · column height from its 3D box · always marked in `derived`
- **provenance** `clean_sources` records which Revit property fed each column, per category

Checked against the failing test questions

walls thicker than 200 mm: **0 → 9** · columns wider than 300 mm: **0 → 1** · structural on Level 2: **0 → 92** · not structural: **801 → 7** · levels: **all elements → 6 levels** · shortest column: **"no length" → 2.793 m⚙**

The 20 columns

Column	Meaning	Unit	Filled*	Main Revit source
`category`	Revit category	text	100%	from the category / tree
`family`	Family name	text	100%	from the category / tree
`type`	Type name (kept as text: "200UB25.4")	text	100%	from the category / tree
`level`	Level the element is on	text	53%	Reference Level, Level
`top_level`	Level it goes up to	text	5%	Top Level, Top Constraint
`mark`	Mark	text	3%	Mark
`material`	Structural material, else material	text	53%	Structural Material
`material_class`	concrete / steel / timber / aluminium / masonry / glass / other	text	53%	⚙ from material
`is_structural`	Revit's Structural flag, else from the category	1 / 0	100%	⚙ from the category, Structural
`usage`	Structural usage (joist, girder...)	text	45%	Structural Usage
`length_m`	Length	m	88%	Length, Total Bar Length
`height_m`	Height (columns: from the 3D view)	m	5%	⚙ from the 3D view (top - bottom of its box), Unconnected Height
`width_mm`	Section / footprint width	mm	47%	b, B
`depth_mm`	Section depth	mm	47%	h, d
`thickness_mm`	Thickness (walls: Revit "Width")	mm	4%	Thickness, Width
`section`	Width x depth, as text	text	43%	from the category / tree
`area_m2`	Area	m²	2%	Area
`volume_m3`	Volume	m³	96%	Volume, Reinforcement Volume
`weight_kg_per_m`	Nominal weight (kg/m and kgf/m)	kg/m	11%	Nominal Weight
`weight_kg`	Weight = kg/m x length (computed)	kg	7%	⚙ weight per metre x length

* share of all 845 elements with a value here (rebar and reinforcement have no level, section or material in this model). ⚙ = computed, not read.

Sample rows (real data, rstbasic)

category	type	level	material_class	is_structural	length_m	height_m	width_mm	depth_mm	thickness_mm	section	area_m2	volume_m3	weight_kg_per_m	weight_kg
Walls	CL_W1	Foundation	concrete	1	9.628	4.7	280	–	280	–	45.252	12.671	–	–
Floors	Insitu Concrete	Level 1	concrete	1	–	–	–	–	550	–	38.46	21.153	–	–
Structural Columns	450 x 450mm	Level 1	concrete	1	–	3⚙	450	450	–	450 x 450 mm	–	0.607	–	–
Structural Columns	200UB25.4	Level 2	steel	1	–	3.1⚙	133	203	–	133 x 203 mm	–	0.01	25.4	–
Structural Framing	200UB25.4	Top of Parapet	steel	1	4.243	–	133	203	–	133 x 203 mm	–	0.013	25.4	107.762⚙
Structural Framing	Timber_50 x200	Level 2	timber	1	1.5	–	50	200	–	50 x 200 mm	–	0.014	–	–
Structural Foundations	600 x 600 x 900mm	Level Lower	concrete	1	0.6	–	600	–	300	–	–	–	–	–
Structural Rebar	Structural Rebar 1	–	–	1	19.89	–	–	–	–	–	–	0.001	–	–

Also verified on Seaport (15,147 elements, imperial units): doors 914.4 mm × 2.134 m (= 3 ft × 7 ft), wall thickness from Revit Width, levels "O-Level 2", "G-Roof"... · rstbasic builds in 58 ms, Seaport in ~1 s · 20 automated checks in test/run_clean.js